DMBI CA3 Content Beyond Syllabus · D15A Group 09

High-Utility Itemset Mining

When the most frequent pattern is not the most valuable one

2 of 20

itemsets shared by the top lists for frequency and for value

1.65×

revenue in the top 20 by utility, against the top 20 by support

77.6%

of high-utility itemsets fall below Apriori's 1% support

Jaiprakash Talreja (54) · Adarsh Singh (52) · Vidushi Singh (53) · Piyush Shukla (49)

Main reference: Gan et al. (2021), A Survey of Utility-Oriented Pattern Mining, IEEE TKDE 33(4)

Apriori counts baskets. It never asks what they are worth

Basket	Items and quantity	Value
B1	Bread × 2, Milk × 1	3
B2	Bread × 1, Milk × 2, Eggs × 1	5
B3	Bread × 1, Eggs × 2	5
B4	Bread × 1, Milk × 1	2
B5	Wine × 2, Cheese × 1	37
B6	Wine × 1, Cheese × 2, Bread × 1	30

Unit prices: Bread 1, Milk 1, Eggs 2, Wine 15, Cheese 7

What Apriori sees

Only whether an item is in the basket. Quantity and price are ignored.

Bread + Milk

in 3 baskets

worth 8

Cheese + Wine

in 2 baskets

worth 66

Utility adds quantity and value to every item

Internal utility

Quantity of an item in a basket. Wine × 2 in B5.

External utility

Value of one unit: price, profit or any weight. Wine = 15.

Itemset utility

Quantity × value, added across all baskets holding the whole itemset.

u(X) = sum over baskets T containing X of [sum over items i in X of quantity(i, T) × value(i)]

Worked example: Cheese + Wine

B5: 2 × 15 + 1 × 7 = 37. B6: 1 × 15 + 2 × 7 = 29. Total utility = 66. With a minimum utility of 30, this itemset is high utility.

Utility breaks the rule that makes Apriori fast



Cheese fails the threshold, yet its superset passes it.

Support: safe to prune

Adding an item can only keep support the same or lower it. An infrequent set can never become frequent, so Apriori drops the whole branch.

Utility: not safe to prune

Adding an item can raise utility or lower it. A low-utility set can grow into a high-utility one, so utility itself cannot be used to prune.

An upper bound restores safe pruning

Item	TWU	Against minimum 30
Wine	67	Keep
Cheese	67	Keep
Bread	45	Keep
Milk	**10**	**Prune safely**
Eggs	**10**	**Prune safely**

TWU(Cheese) = 37 + 30 = 67: the full value of B5 and B6.

Transaction-weighted utility

Add the whole basket value for every basket holding the itemset. It never underestimates the utility of any superset.

Why this is not enough

TWU is loose, so many candidates survive. Two-Phase (Liu et al., 2005) must then scan again to get exact utilities.

HUI-Miner keeps exact values in a utility-list

Cheese

Basket	iutil	rutil
B5	7	30
B6	14	15

join

Wine

Basket	iutil	rutil
B5	30	0
B6	15	0

Cheese + Wine

Basket	iutil	rutil
B5	37	0
B6	29	0

1. Scan once

Compute TWU, drop weak items

2. Build lists

One utility-list per item

3. Join

Extend itemsets by matching baskets

4. Prune

Stop if iutil + rutil is below minimum

The pruning rule

Cheese: iutil 21 + rutil 45 = 66, at least 30, so extend it. The bound never underestimates, so no high-utility itemset is lost.

We built it and tested it on real retail baskets

541,909

raw invoice lines

19,636

invoices after cleaning

3,786

products

£9,873,955

total revenue

Cleaning

Removed 3,188 cancelled sales as matched pairs, plus postage, fees and other non-products.

Utility here means revenue

Quantity × unit price. Profit is not in the data, so high utility means high revenue, not high margin.

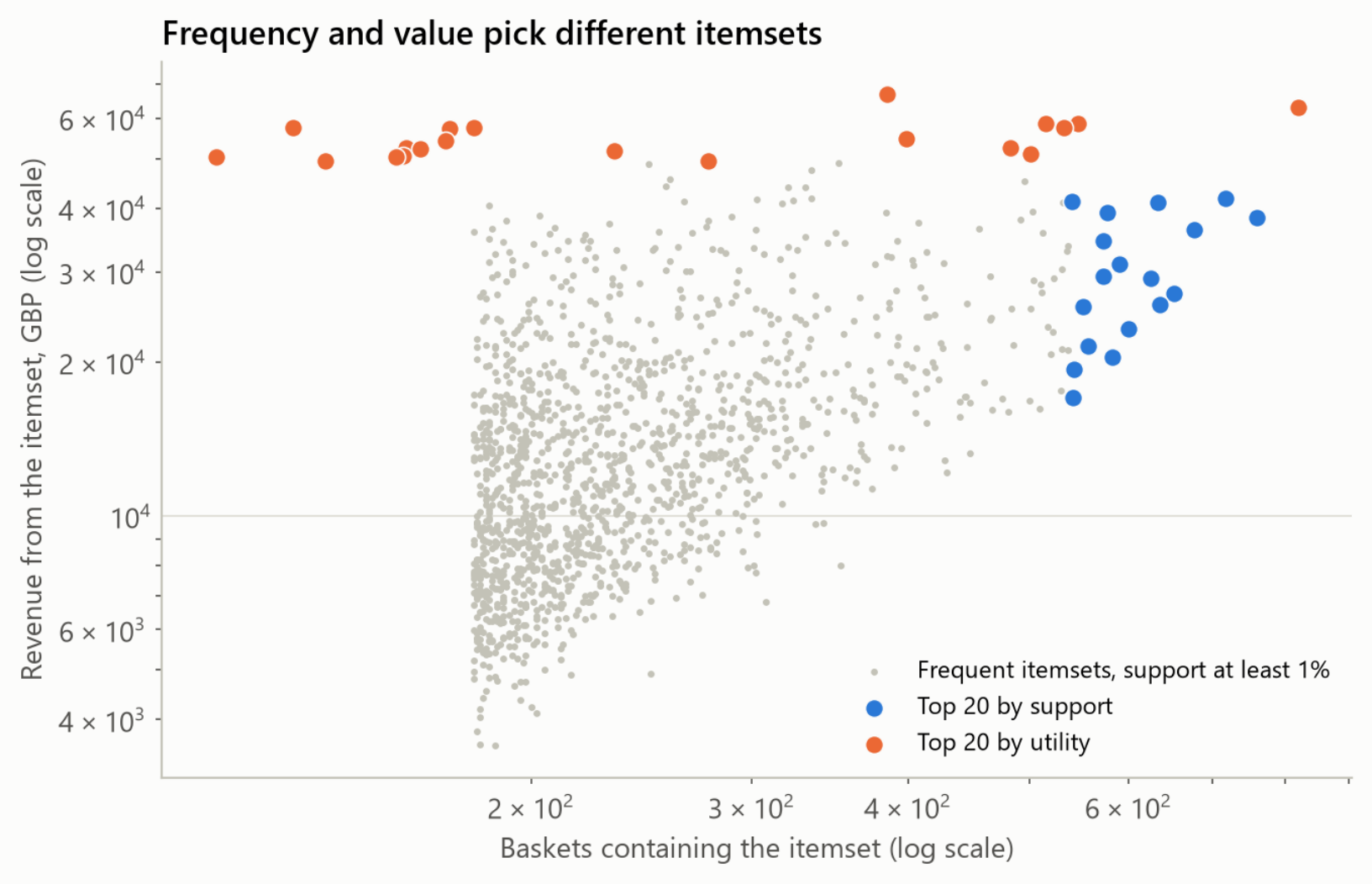
Scope

Top 300 products (54.7% of revenue), up to 3 items, minimum utility £24,685 (0.25% of revenue). Exact within that scope.

Checked

Matched exhaustive search on 300 random databases. Invoices reach 1,106 lines, which is why scope is limited.

Frequency and value pick different itemsets



2 of 20

itemsets are in both top lists

1.65×

revenue: £1,096,425 against £664,360

77.6%

of 777 high-utility itemsets are below 1% support

The two lists name different products to promote

Top by support (what Apriori favours)

Itemset	Baskets	Revenue
Jumbo Bag Pink Polkadot + Jumbo Bag Red Retrospot	819	£63,091
Green Regency Teacup And Saucer + Roses Regency Teacup And Saucer	759	£38,337
Jumbo Bag Red Retrospot + Jumbo Storage Bag Suki	717	£41,869

Top by utility (what earns most)

Itemset	Baskets	Revenue
Party Bunting + Regency Cakestand 3 Tier	385	£66,896
Jumbo Bag Pink Polkadot + Jumbo Bag Red Retrospot	819	£63,091
Regency Cakestand 3 Tier + Roses Regency Teacup And Saucer	515	£58,708

825

high-utility itemsets found at 0.25% of revenue

1,448,423

utility-lists built to find them

36,022

joins skipped by the FHM pair check

16,202

branches stopped by iutil + rutil

Where it helps, and where it falls short

Applications

- Cross-marketing and promotion planning
- E-commerce
- Finance
- Medical and biomedical analysis

Advantages

- Finds valuable patterns that frequency misses
- Uses quantity and value, which Apriori discards
- Works with profit, price or any numeric weight
- Exact: pruning loses no high-utility itemset

Limitations

- Needs a reliable value per item
- Long itemsets from bulk orders can dominate
- Large baskets weaken pruning
- A minimum utility is harder to set than a support

Frequent is not the same as valuable

High-utility itemset mining extends Module V by measuring worth, not just counts. On real baskets it changed 18 of the top 20 answers.

References

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Thank you. We are open for your questions.